

Abhay Korlapati

(310) 894-9312 — akkorlapati@gmail.com — github.com/abhay784 — linkedin.com/in/abhay-korlapati-954a63360

EDUCATION

University of California, San Diego

Jun 2028

B.S. Computer Science

- Relevant Coursework: Data Structures & Algorithms, Systems Programming, Discrete Math, Linear Algebra
- CyberStart America Finalist (2023); CyberPatriots Team Captain (2017–2024)

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL

Frameworks/Libraries: NumPy, Pandas, Kaggle, Next.js, Scikit-Learn, TensorFlow

Tools/Platforms: Docker, MySQL, Supabase, Firebase

EXPERIENCE

Engineers for Exploration (UC San Diego)

Oct 2025 – Present

Research Assistant, Acoustic Bird Project

La Jolla, CA

- Simulate bird calls and environmental acoustics in Unreal Engine to generate controlled test audio for a wildlife monitoring pipeline
- Model sound propagation (distance attenuation, occlusion, reverb) to match real field conditions and evaluate microphone array performance
- Compare synthetic vs. in-field recordings to explain why model accuracy drops on real data and isolate failure modes like background noise and interference

Triton NeuroTech (UC San Diego)

Sep 2025 – Present

Member, EMG Prosthetics Team

La Jolla, CA

- Utilized MindRove EMG armband and Python SDK to acquire and preprocess myoelectric signals for real-time prosthetic control.
- Implemented band-pass and Butterworth filters to remove motion artifacts and noise, improving EMG signal quality.
- Trained a TensorFlow LSTM gesture classifier and optimized it with GWO and TPR/FPR-based threshold tuning.

PROJECTS

BrokerBot — Home Project Quote Negotiator

Oct 2025–Present

- Built an AI “buyer’s rep” using a Claude agent that contacted 20+ local contractors across plumbing, electrical, tile, and GC trades, automating 100% of quote outreach via email/SMS through a relay identity.
- Parsed 50–100 unstructured vendor replies into structured quote objects (price, scope, permits, warranty, start date), enabling direct comparison and highlighting 5–7 missing scope items per project on average.
- Auto-generated negotiation counter-messages that used competing bids as leverage and produced a final summary of who to hire and what to confirm before paying a deposit

Voice-Based Gender Classification

- Built a speech classification pipeline to predict speaker gender from short audio samples using classical ML (no deep learning) for interpretability and low compute cost
- Cleaned and normalized 3K+ raw WAV clips, trimmed silence, and converted each sample into spectrogram and MFCC feature vectors using Python and NumPy
- Engineered per-sample statistical features (pitch range, formant energy, spectral centroid) and assembled a training matrix with pandas for model ingestion
- Trained and evaluated a logistic regression classifier in scikit-learn with regularization and an 80/20 train/validation split, reaching 92% accuracy on held-out validation data
- Plotted confusion matrix and error cases in Matplotlib to identify failure modes such as background noise and overlapping voices

Legend of Dreamy — 2D RPG Game

March 2024–June 2024

- Built a 2D RPG with custom combat engine, branching narrative, enemy AI, and boss logic implemented in Java
- Integrated 25+ original sprites, multiple soundtracks, animation frames, and state-based rendering loops for attacks, movement, and triggers
- Implemented inventory, dialogue, and progression systems without a game engine framework, focusing on game loop timing and collision handling from scratch